Robótica de Manipulação: 8^{th} Problems Class

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Introduction. Modelling vs. Formulation

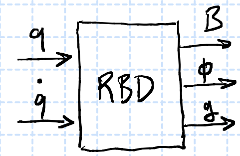
Rigid Body Dynamics

- Lagrange-Euler
- Newton-Euler
- etc...

Equations of motion:

$$B(q)\,\ddot{q} + \underbrace{\overbrace{C(q,\dot{q})\,\dot{q}}^{\phi(q,\dot{q})} + g(q)}_{n(q,\dot{q})} = \tau$$

In the project, construct a block with B, ϕ, g.



Lagrange-Euler,

$$\mathcal{L} = T(q,\dot{q}) - U(q)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k}\right) - \frac{\partial \mathcal{L}}{\partial q_k} = \tau_k \;,\; k=1,\dots,n$$

Mass Matrix:

$$B = \sum_{i=1}^{n} \underbrace{m_{\ell_i}\, J_P^{(\ell_i)T} J_P^{(\ell_i)}}_{B_P^{(\ell_i)}} + \underbrace{J_O^{(\ell_i)T} \left(R_i\, I_{\ell_i}^i\, R_i^T\right) J_O^{(\ell_i)}}_{B_O^{(\ell_i)}} = \sum_{i=1}^{n} B_P^{(\ell_i)} + B_O^{(\ell_i)}$$

Coriolis/Centrifugal matrix:

$$c_{ij} = \sum_{k=1}^{n} h_{ijk}\,\dot{q}_k \;,\quad h_{ijk} = \frac{\partial B_{ij}}{\partial q_k} - \frac{1}{2}\frac{\partial B_{jk}}{\partial q_i}$$

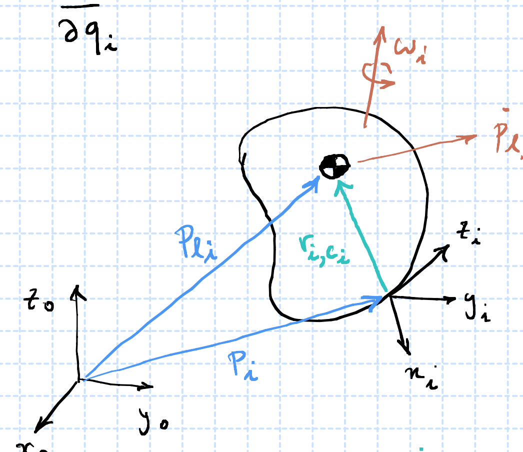
Gravity vector:

$$g_i = -\sum_{j=1}^{n} m_{\ell_j}\, g_0^T \underbrace{J_{P_i}^{(\ell_j)}}_{= \frac{\partial p_{\ell_j}}{\partial q_i}} = \frac{\partial U}{\partial q_i}$$

$$J_P^{(\ell_i)} = \frac{\partial p_{\ell_i}}{\partial q}$$

$$J_O^{(\ell_i)} = \begin{pmatrix} J_{O_1}^{(\ell_i)} & \dots & J_{O_i}^{(\ell_i)} & 0 & \dots & 0 \end{pmatrix}$$

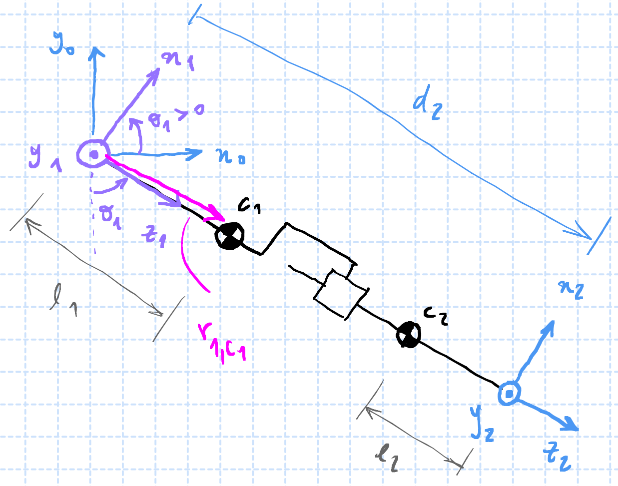
$$J_{O_j}^{(\ell_i)} = \begin{cases} z_{j-1} \;, & j \text{ is revolute} \wedge j \le i \\ 0 \;, & \text{otherwise} \end{cases}$$



$$p_{\ell_i} = p_i + R_i\, r_{i,c_i}^i$$

$$\begin{pmatrix} \dot{p}_{\ell_i} \\ \omega_i \end{pmatrix} = \begin{pmatrix} J_P^{(\ell_i)} \\ \hline J_O^{(\ell_i)} \end{pmatrix} \dot{q}$$

1.



(a) direct kinematics

$$R_1^0 = (x_1^0\ y_1^0\ z_1^0) = \begin{pmatrix} c_1 & 0 & s_1 \\ s_1 & 0 & -c_1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$R_2^0 = R_1^0 \underbrace{R_2^1}_{I} = R_1^0$$

$$p_{l_1}^0 = p_1^0 + R_1^0 r^1_{1,c_1}$$

$$p_{l_2}^0 = p_2^0 + R_2^0 r^2_{2,c_2}$$

$$p_{l_1} = l_1 z_1 = \begin{pmatrix} l_1 s_1 \\ -l_1 c_1 \\ 0 \end{pmatrix}$$

$$p_{l_2} = (d_2 - l_2) z_1 = \begin{pmatrix} (d_2-l_2) s_1 \\ -(d_2-l_2) c_1 \\ 0 \end{pmatrix}$$

(b) differential kinematics

$$J_p^{(l_1)} = \frac{\partial p_{l_1}}{\partial q} = \left(\frac{\partial p_{l_1}}{\partial \theta_1} \,\Big|\, \frac{\partial p_{l_1}}{\partial d_2} \right) = \begin{pmatrix} l_1 c_1 & 0 \\ l_1 s_1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$J_p^{(l_2)} = \frac{\partial p_{l_2}}{\partial q} = \begin{pmatrix} (d_2-l_2) c_1 & s_1 \\ (d_2-l_2) s_1 & -c_1 \\ 0 & 0 \end{pmatrix}$$

$$J_o^{(l_1)} = \left(z_0^0 \,|\, 0 \right) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$J_o^{(l_2)} = J_o^{(l_1)}$$

(c) Mass matrix, $B = \sum_{i=1}^{n} \underbrace{m_{l_i} J_p^{(l_i)T} J_p^{(l_i)}}_{B_p^{(l_i)}} + \underbrace{J_o^{(l_i)T} (R_i I_{l_i}^i R_i^T) J_o^{(l_i)}}_{B_o^{(l_i)}} = \sum_{i=1}^{n} B_p^{(l_i)} + B_o^{(l_i)}$

$$B_p^{(l_1)} = m_{l_1} J_p^{(l_1)T} J_p^{(l_1)} = m_1 \begin{pmatrix} l_1 c_1 & l_1 s_1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 c_1 & 0 \\ l_1 s_1 & 0 \\ 0 & 0 \end{pmatrix} = m_1 \begin{pmatrix} l_1^2 (c_1^2 + s_1^2) & 0 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} m_1 l_1^2 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$B_p^{(l_2)} = m_{l_2} J_p^{(l_2)T} J_p^{(l_2)} = m_2 \begin{pmatrix} (d_2-l_2) c_1 & (d_2-l_2) s_1 & 0 \\ s_1 & -c_1 & 0 \end{pmatrix} \begin{pmatrix} (d_2-l_2) c_1 & s_1 \\ (d_2-l_2) s_1 & -c_1 \\ 0 & 0 \end{pmatrix} =$$

$$J_p^{(l_1)} = \begin{pmatrix} l_1 c_1 & 0 \\ l_1 s_1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$J_p^{(l_2)} = \begin{pmatrix} (d_2-l_2) c_1 & s_1 \\ (d_2-l_2) s_1 & -c_1 \\ 0 & 0 \end{pmatrix}$$

$$= m_2 \begin{pmatrix} (d_2-l_2)^2 & (d_2-l_2) c_1 s_1 - (d_2-l_2) c_1 s_1 \\ \text{sym.} & s_1^2 + c_1^2 \end{pmatrix} =$$

$$= \begin{pmatrix} m_2 (d_2-l_2)^2 & 0 \\ 0 & m_2 \end{pmatrix}.$$

$$B_0^{(\ell_1)} = y_0^{(\ell_1)T} \underbrace{\left(R_1^0 I_{\ell_1}^1 R_0^1\right)}_{= I_{\ell_1}^0} y_0^{(\ell_1)} = \qquad \underbrace{y_0^{(\ell_i)T}\left(R_i I_{\ell_i}^i R_i^T\right) y_0^{(\ell_i)}}_{B_0^{(\ell_i)}}$$

(second order tensor transformation)

$$= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \underbrace{\begin{pmatrix} * & * & * \\ * & * & * \\ * & * & I_z^0 \end{pmatrix}_{\ell_1}}_{I_{\ell_1}^0} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$I_{\ell_1}^1 = \begin{pmatrix} I_x^1 & I_{xy}^1 & I_{xz}^1 \\ & I_y^1 & I_{yz}^1 \\ \text{sym.} & & I_z^1 \end{pmatrix}_{\ell_1}$$

$$y_0^{(\ell_1)} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$I_{\ell_1}^0 = R_1^0 I_{\ell_1}^1 R_0^1 = \underbrace{\begin{pmatrix} c_1 & 0 & s_1 \\ s_1 & 0 & -c_1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} & * & \\ & I_y^1 & \\ & * & \end{pmatrix}_{\ell_1}}_{\begin{pmatrix} & & \\ & & \\ * & I_z^0 & * \end{pmatrix}} \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix}$$

$$R_1^0 = \begin{pmatrix} c_1 & 0 & s_1 \\ s_1 & 0 & -c_1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\boxed{I_z^0 = I_y^1}$$

$$B_0^{(\ell_1)} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} * & * & * \\ * & * & * \\ * & * & I_y^1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} I_{y,\ell_1}^1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$B_0^{(\ell_2)} = \begin{pmatrix} I_{y,\ell_2}^2 & 0 \\ 0 & 0 \end{pmatrix} \qquad \text{(by inspection)}$$

$$B = B_p^{(\ell_1)} + B_p^{(\ell_2)} + B_0^{(\ell_1)} + B_0^{(\ell_2)} = \begin{pmatrix} m_1 \ell_1^2 + m_2 (d_2-\ell_2)^2 + I_{y,\ell_1}^1 + I_{y,\ell_2}^2 & 0 \\ 0 & m_2 \end{pmatrix}$$

(d) Coriolis / centrifugal matrix, $c_{ij} = \sum_{k=1}^{n} h_{ijk} \dot{q}_k$, $h_{ijk} = \dfrac{\partial B_{ij}}{\partial q_k} - \dfrac{1}{2} \dfrac{\partial B_{jk}}{\partial q_i}$

$$h_{ij1} = \begin{pmatrix} \cancel{h_{111}}^{0} & \cancel{h_{121}}^{0} \\ \underline{h_{211}} & \cancel{h_{221}}^{0} \end{pmatrix}, \quad h_{ij2} = \begin{pmatrix} \underline{h_{112}} & \cancel{h_{122}}^{0} \\ \cancel{h_{212}}_{0} & \cancel{h_{222}}_{0} \end{pmatrix}$$

$$h_{111} = \frac{\partial B_{11}}{\partial \theta_1} - \frac{1}{2} \frac{\partial B_{11}}{\partial \theta_1} = 0$$

$$h_{121} = \frac{\partial B_{12}}{\partial \theta_1} - \frac{1}{2} \frac{\partial B_{21}}{\partial \theta_1} = 0$$

$$h_{211} = \cancel{\frac{\partial B_{21}}{\partial \theta_1}}^{0} - \frac{1}{2} \frac{\partial B_{11}}{\partial d_2} = -\frac{1}{2}\left[2 m_2 (d_2-\ell_2)\right] = -m_2 (d_2-\ell_2)$$

$$h_{221} = \frac{\partial B_{22}}{\partial \theta_1} - \frac{1}{2} \frac{\partial B_{21}}{\partial d_2} = 0$$

$$B(d_2) = \begin{pmatrix} m_1 \ell_1^2 + m_2 (d_2-\ell_2)^2 + I_{y,\ell_1}^1 + I_{y,\ell_2}^2 & 0 \\ 0 & m_2 \end{pmatrix}$$

$$h_{112} = \frac{\partial B_{11}}{\partial d_2} - \frac{1}{2}\cancelto{0}{\frac{\partial B_{12}}{\partial \theta_1}} = 2m_2(d_2-\ell_2)$$

$$B(d_2) = \begin{pmatrix} m_1\ell_1^2 + m_2(d_2-\ell_2)^2 + I^1_{y,\ell_1} + I^2_{y,\ell_2} & 0 \\ 0 & m_2 \end{pmatrix}$$

$$h_{122} = \frac{\partial B_{12}}{\partial d_2} - \frac{1}{2}\frac{\partial B_{22}}{\theta_1} = 0$$

$$h_{ijk} = \frac{\partial B_{ij}}{\partial q_k} - \frac{1}{2}\frac{\partial B_{jk}}{\partial q_i}$$

$$h_{212} = \frac{\partial B_{21}}{\partial d_2} - \frac{1}{2}\frac{\partial B_{12}}{\partial d_2} = 0$$

$$h_{222} = 0$$

$$h_{ij1} = \begin{pmatrix} 0 & 0 \\ -m_2(d_2-\ell_2) & 0 \end{pmatrix}, \quad h_{ij2} = \begin{pmatrix} 2m_2(d_2-\ell_2) & 0 \\ 0 & 0 \end{pmatrix}$$

$$C_{11} = \cancelto{0}{h_{111}}\dot\theta_1 + h_{112}\dot d_2 = 2m_2(d_2-\ell_2)\dot d_2$$

$$C_{12} = \cancelto{0}{h_{121}}\dot\theta_1 + \cancelto{0}{h_{122}}\dot d_2 = 0$$

$$C_{ij} = \sum_{k=1}^{n} h_{ijk}\dot q_k$$

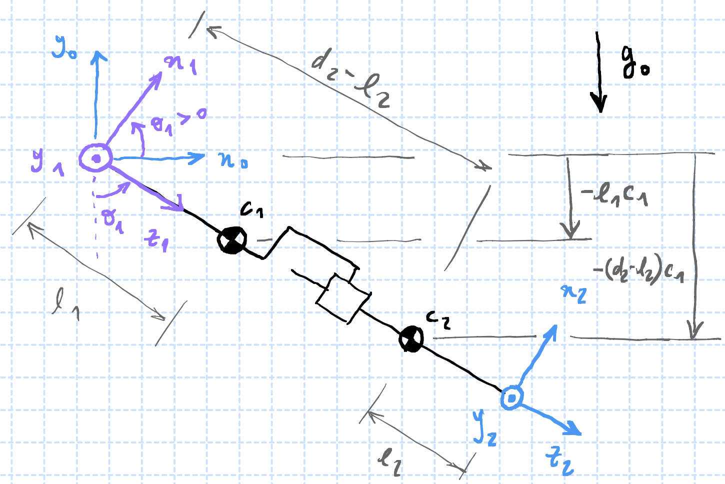
$$C_{21} = h_{211}\dot\theta_1 + \cancelto{0}{h_{212}}\dot d_2 = -m_2(d_2-\ell_2)\dot\theta_1$$

$$C_{22} = 0$$

$$\phi = C\dot q = \begin{pmatrix} 2m_2(d_2-\ell_2)\dot d_2 & 0 \\ -m_2(d_2-\ell_2)\dot\theta_1 & 0 \end{pmatrix}\begin{pmatrix} \dot\theta_1 \\ \dot d_2 \end{pmatrix} = \begin{pmatrix} 2m_2(d_2-\ell_2)\dot\theta_1\dot d_2 \\ -m_2(d_2-\ell_2)\dot\theta_1^2 \end{pmatrix}$$

← Coriolis
← centrifugal forces

(e) gravity vector, $g_i = -\sum_{j=1}^{n} m_{\ell_i} g_0^T y_{P_i}^{(\ell_i)} = \frac{\partial U}{\partial q_i}$ $\qquad U = mgh$



$$U = -m_1|g|\ell_1 c_1 - m_2|g|(d_2-\ell_2)c_1$$

$$\frac{\partial U}{\partial \theta_1} = m_1|g|\ell_1 s_1 + m_2|g|(d_2-\ell_2)s_1$$

$$\frac{\partial U}{\partial d_2} = -m_2|g|c_1$$

$$g = \begin{pmatrix} [m_1\ell_1 + m_2(d_2-\ell_2)]|g|s_1 \\ -m_2|g|c_1 \end{pmatrix}.$$

$$g_0 \approx \begin{pmatrix} 0 \\ -9{,}81 \\ 0 \end{pmatrix}[m/s^2], \quad g_0 = \begin{pmatrix} 0 \\ -|g| \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} m_1\ell_1^2 + m_2(d_2-\ell_2)^2 + I^1_{y,\ell_1} + I^2_{y,\ell_2} & 0 \\ 0 & m_2 \end{pmatrix}\begin{pmatrix} \ddot\theta_1 \\ \ddot d_2 \end{pmatrix} + \begin{pmatrix} 2m_2(d_2-\ell_2)\dot\theta_1\dot d_2 \\ -m_2(d_2-\ell_2)\dot\theta_1^2 \end{pmatrix} +$$

$$+ \begin{pmatrix} [m_1\ell_1 + m_2(d_2-\ell_2)]|g|s_1 \\ -m_2|g|c_1 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ f_1 \end{pmatrix}$$